

How We Test the Baselines

A why-and-how guide to the SPARC benchmark metrics

1 The one idea to keep in mind

Every generator we compare — the copula, GMMNet, EV-SDG, and SPARC — produces synthetic charging sessions, each a triple

session = (A, D, E) : A = arrival hour, D = plug-in duration (h), E = energy (kWh).

The crucial difference is *bookkeeping*. The three baselines emit an anonymous **bag of sessions**: thousands of (A, D, E) triples with no label saying *which driver* produced each one. SPARC instead emits sessions **tagged by user**, because it fits one model per user.

That single difference is why we test along **two axes**, not one:

1. **Marginal fidelity** (Axis 1): do the overall distributions of A, D, E match reality? This is the conventional test and it ignores user identity.
2. **Per-user and joint structure** (Axis 2): does each *individual* behave realistically, and does the fleet’s combined load come out right? This is where identity matters.

The headline finding of the benchmark is that a model can *ace Axis 1 and fail Axis 2*. The copula reproduces the population histograms almost perfectly, yet it manufactures a fleet of near-identical “average” drivers. The rest of this note explains each metric behind these two axes.

2 Axis 1 — Marginal fidelity via KL divergence

2.1 What a marginal is

A *marginal* is just the distribution of one variable, ignoring the others: $p(A)$ is the histogram of arrival hours over *all* sessions from *all* users; likewise $p(D)$ and $p(E)$. It answers “how often do sessions start at 18:00?” but says nothing about *who* started them.

2.2 How we score it: Kullback–Leibler (KL) divergence

We measure how far a model’s marginal q is from the real marginal p using the forward KL divergence,

$$D_{\text{KL}}(p \parallel q) = \sum_k p_k \ln \frac{p_k}{q_k} \quad (\text{nats}),$$

where k indexes histogram bins (24 hourly bins for A ; 30 bins for D and E). Read it as: “*if reality is p , how many nats of information are lost by pretending the world is q ?*” Properties worth internalising:

- $D_{\text{KL}} = 0$ exactly when the two histograms are identical; larger means worse.

- It is measured in *nats* (natural-log units). In our work $D_{\text{KL}} < 0.02$ is “excellent,” ≈ 0.1 is visibly off, and > 1 is a different distribution.
- We use the *forward* direction $D_{\text{KL}}(\text{real}||\text{model})$, which heavily penalises a model that puts (almost) no probability where reality has mass.

A worked micro-example. Suppose real arrivals fall in three bins with probabilities $p = (0.5, 0.3, 0.2)$ and a model predicts $q = (0.4, 0.4, 0.2)$. Then

$$D_{\text{KL}} = 0.5 \ln \frac{0.5}{0.4} + 0.3 \ln \frac{0.3}{0.4} + 0.2 \ln \frac{0.2}{0.2} = 0.112 - 0.086 + 0 = 0.025 \text{ nats.}$$

Small, because the histograms are close. This is exactly the scale of SPARC’s arrival-hour scores.

2.3 What Axis 1 shows

The copula reproduces the pooled histograms almost exactly ($D_{\text{KL}} \leq 0.001$ on the large pools) because it is *built* from the empirical marginals. SPARC is a close second everywhere. EV-SDG is the outlier: it caps every session at a 24 h horizon, so on residential data with overnight sessions (real median 12 h, tail beyond 100 h) its duration marginal has almost no overlap with reality and the KL explodes to ~ 18 nats. **On this axis alone, one might simply prefer the copula** — which is precisely the trap Axis 2 springs.

3 Axis 2 — Per-user structure

Here we stop pooling and look at each user. Three terms do the work: *mean plug-in duration*, *circular mean arrival hour*, and the *across-user standard deviation*.

3.1 Mean plug-in duration (per user)

The easy one. For user u with sessions $\{D_1, \dots, D_{n_u}\}$, the per-user mean duration is the ordinary average $\bar{D}_u = \frac{1}{n_u} \sum_i D_i$. It summarises “how long this particular driver typically stays plugged in.” One number per user.

3.2 Circular mean arrival hour (per user) — and why “circular”

Arrival hour lives on a **clock**, not a ruler: hour 23:00 and hour 01:00 are two hours apart, not twenty-two. If you average them with the ordinary (linear) mean you get 12:00 — the middle of the day — which is absurd for a driver who always arrives late at night.

The fix is to treat each hour as an **angle**, average the unit vectors, and convert back:

$$\theta_i = \frac{2\pi A_i}{24}, \quad \bar{A}_u = \frac{24}{2\pi} \text{atan2}\left(\sum_i \sin \theta_i, \sum_i \cos \theta_i\right) \pmod{24}.$$

For the three arrivals $\{22, 23, 1\}$ this gives $\bar{A}_u = 23.3$ h (correct), versus the linear mean 15.3 h (wrong). Every per-user arrival summary in the benchmark uses this circular mean.

Three arrivals at 22:00, 23:00, 01:00

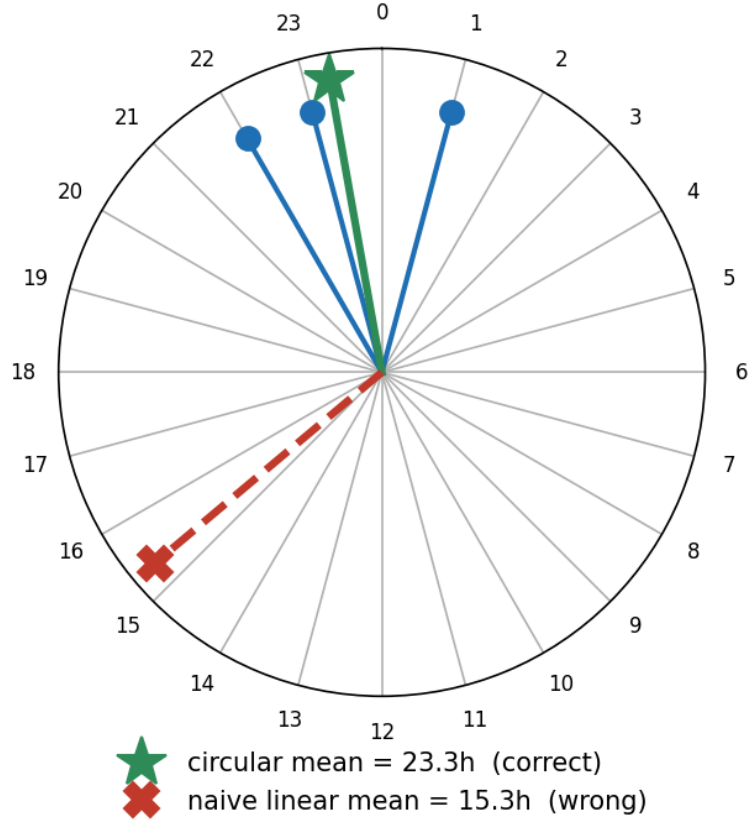


Figure 1: Why arrival hour needs a *circular* mean. Three arrivals at 22:00, 23:00, 01:00 cluster near midnight. The circular mean (green star, 23.3h) sits inside the cluster; the naive linear mean (red X, 15.3h) lands on the opposite side of the clock — completely wrong.

3.3 Across-user standard deviation σ_A, σ_D — the heterogeneity signature

Now the key quantity. We have one number per user (its $\bar{A}_u$, or its $\bar{D}_u$). The **across-user standard deviation** is the spread of *those per-user numbers* across the whole fleet:

$$\sigma_D = \text{std}_u(\bar{D}_u), \quad \sigma_A = (\text{circular}) \text{std}_u(\bar{A}_u).$$

It answers: “*do different drivers actually behave differently?*”

- **Large** $\sigma \Rightarrow$ a diverse fleet: some drivers are morning commuters, others overnight chargers. This is property **P1 (per-user heterogeneity)**.
- **Small** $\sigma \Rightarrow$ everyone looks like the population average: a fleet of clones.

(For arrival we again use the *circular* standard deviation, $\sigma_A = \frac{24}{2\pi} \sqrt{-2 \ln \bar{R}}$, where $\bar{R}$ is the length of the mean unit vector of the per-user angles; the ruler-vs-clock caveat applies here too.)

3.4 The pseudo-user trick: scoring baselines that have no users

Here is the subtlety. The baselines emit an anonymous bag — so how do we compute a *per-user* statistic for them? We give them the fairest possible benefit of the doubt: we **split their pooled sessions into K pseudo-users**, where K and the per-user session counts match the real dataset, then compute σ exactly as above. Averaging over many random splits removes luck.

Crucially, this isolates the effect of losing identity. Figure 2 makes the point with the *real* Norwegian sessions: we take the identical multiset of sessions and only change the *assignment* of sessions to users.

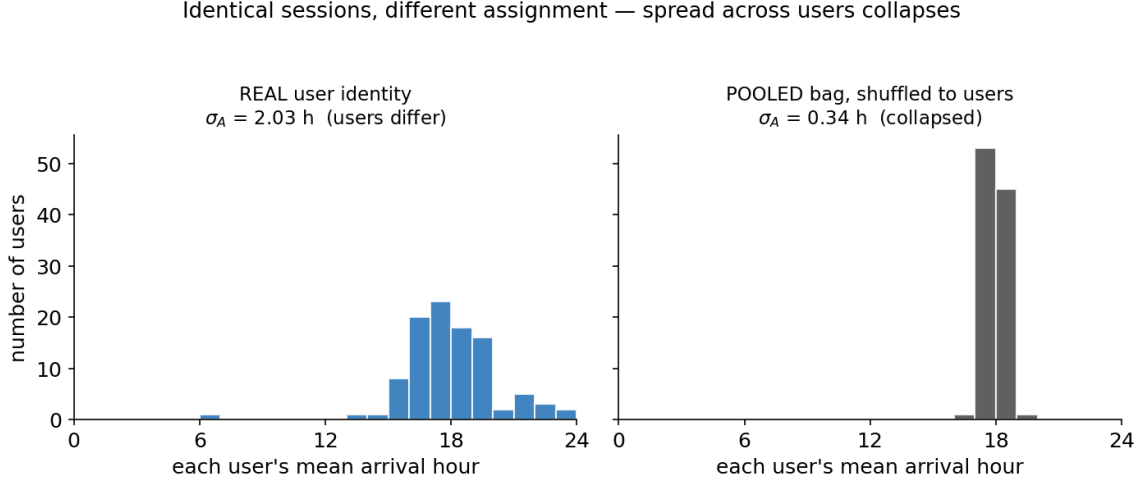


Figure 2: The collapse, on real Norway data. *Left*: with the true user labels, per-user mean arrival hours are spread out ($\sigma_A = 2.03$ h) — a genuinely diverse fleet. *Right*: the *same sessions* randomly re-assigned to users (what a pooled generator implicitly assumes) pile every user’s mean near the population average ($\sigma_A = 0.34$ h). Nothing about the marginals changed; only identity was destroyed.

This is why marginal fidelity and heterogeneity are *independent* tests. Figure 3 shows two fleets with an **identical aggregate marginal** but opposite internal composition.

3.5 What Axis 2 shows

SPARC reproduces the real across-user spread to within its seed variation on every dataset (e.g. Norway $\sigma_D = 5.46 \pm 0.13$ h against a real 5.66 h). The three pooled baselines collapse it three- to six-fold. They pass Axis 1 and fail Axis 2 — they get the crowd right and every individual wrong.

4 Axis 2b — The joint load shape

A second structural test that marginals cannot see is the fleet’s **aggregate daily load profile**. Spreading each session’s power E_i/D_i over the clock hours it occupies gives

$$L(h) = \sum_i \frac{E_i}{D_i} |[A_i, A_i + D_i] \cap [h, h + 1]|, \quad h = 0, \dots, 23.$$

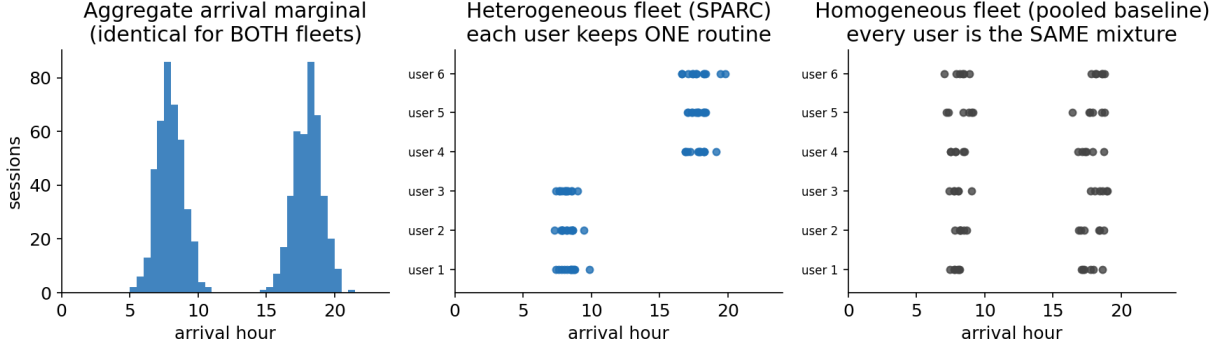


Figure 3: Same marginal, different structure. *Left*: both fleets share the identical bimodal arrival histogram. *Middle*: a heterogeneous fleet (SPARC) — each user keeps one routine (some morning, some evening). *Right*: a homogeneous fleet (pooled baseline) — every user is the same morning+evening mixture. Only the per-user view distinguishes them.

This is a *joint* functional: it depends on how A , D , and E are **paired within a session**, not just on their separate histograms. Figure 4 shows two session sets with *identical* A , D , E marginals but opposite A – D pairings; their load curves are completely different. We score the normalised load shape with the same KL divergence, D_{KL}^L . SPARC attains the lowest D_{KL}^L on all five datasets.

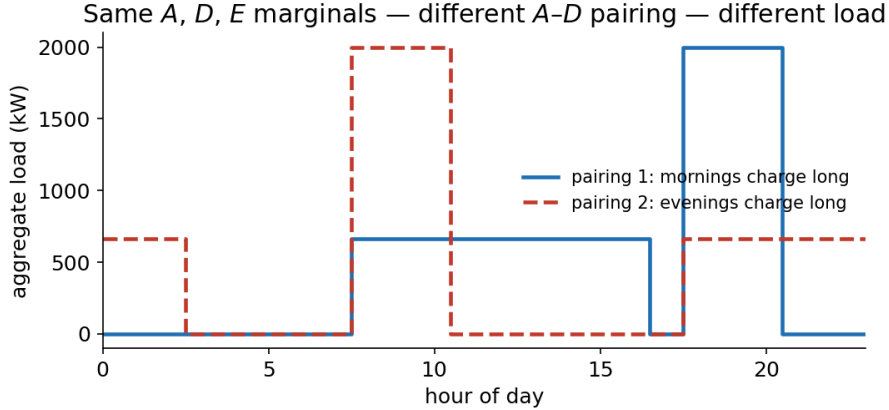


Figure 4: The load is a joint quantity. Both populations have the same A , D , and E marginals; they differ only in *which* arrivals get the long durations. The resulting 24 h load profiles peak at different times and heights — so matching marginals does not guarantee matching grid impact.

5 Why the error bars (the seed sweep)

A single run could be lucky. To show the results are stable we report **mean $\pm$ standard deviation**. For SPARC the spread is taken over **five independent simulation seeds**; for the baselines it is the estimator’s own variability (bootstrap resampling for the KL scores, and the random pseudo-user partitions for σ). SPARC’s seed spread is tiny (e.g. σ_A std ≤ 0.13 h), and the gap to the baselines is many standard deviations wide — so the separation is decisive, not noise.

6 How to read the two benchmark tables

Symbol	Table	Plain-English meaning
A, D, E (KL)	marginals	how far each overall histogram is from real (lower better)
σ_A	structure	spread of per-user <i>arrival</i> routines (closer to Real better)
σ_D	structure	spread of per-user <i>durations</i> (closer to Real better)
D_{KL}^L	structure	error in the 24 h fleet load shape (lower better)

The two tables are meant to be read *together*. The marginal table asks “does the crowd look right?” and the copula wins it. The structure table asks “does each driver look right, and does the fleet load come out right?” and only SPARC matches reality. Matching marginals is necessary but not sufficient: a generator can be marginally perfect yet produce a fleet of identical users, misrepresenting exactly the between-user diversity that drives coincident peak demand, transformer sizing, and managed-charging headroom.

One-line takeaway. SPARC is the only method that simultaneously matches the marginals (Axis 1), the per-user heterogeneity (Axis 2), and the joint load shape (Axis 2b) — and it does so from only tens of sessions per user, without per-context tuning.